Rock Paper Scissor in C

Rock Paper Scissor (which is also called **Stone Paper Scissor**) is a hand game and played between two people, in which each player simultaneously forms one of three shapes. The winner of the game is decided as per the below rules:

- Rock vs Paper -> Paper wins.
- Rock vs Scissor -> Rock wins.
- Paper vs Scissor -> Scissor wins.

In this game, the user will be asked to make choice and according to the choice of user and computer and then the result will be displayed along with the choices of both computer and user.

Approach: Below is the functionality that needed to be implemented in the program:

main() function:

- It consists of the declaration of the variables.
- printf() and scanf() functions for displaying the content and taking input from the user. It also contains two predefined functions:
    - srand() and rand() which are used to generate random numbers in the range **[0, RAND_MAX)** and **srand()** especially will help to generate a random number at each time.
    - Take modulo of random number generated with 100 to make its range between (0 and 100).
    - As the range is up to 100 only, the distribution among all the options i.e., **stone**, **paper**, and **scissors** is equal as all of them have an equal probability of coming.

Note: *This random number will decide the choice of computer as:*

- *If the number is between 0-33 then the choice will be **Stone**.*
- *If the number is between 33-66 then the choice will be **Paper**.*
- *If the number is between 66-100 then the choice will be **Scissors**.*

game() function: This function consists of if-else statements that will compare the choice of user and computer. If the user wins then it will return **1**. Otherwise, if the computer wins then it will return **0**. If it is a tie, it will return **-1**.

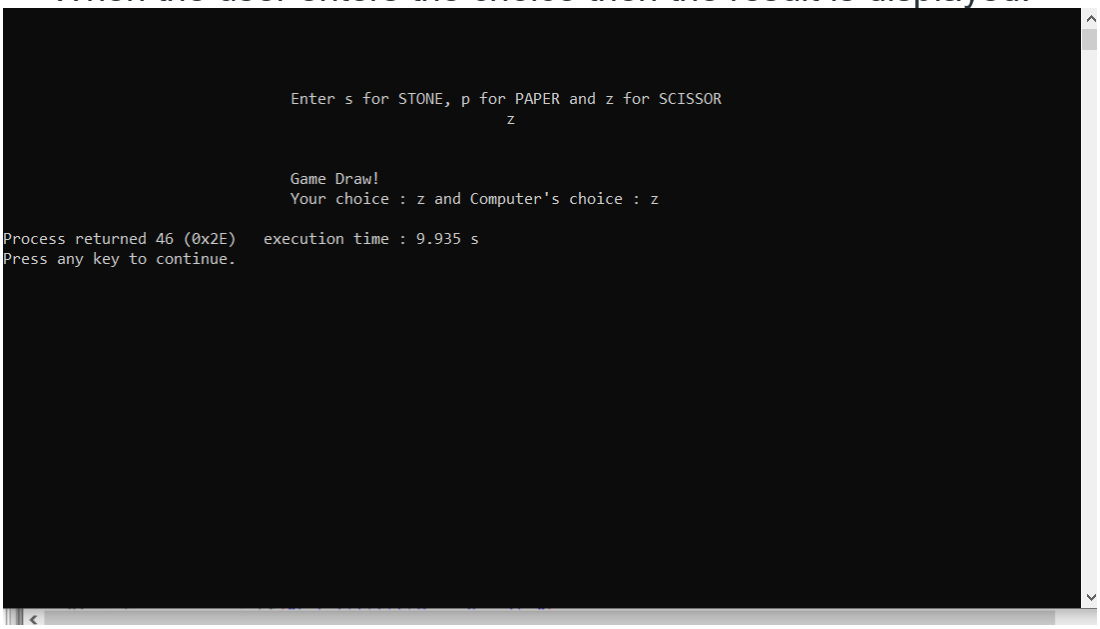
Below is the implementation of the above approach:

```
// C program for the above approach
#include <math.h>
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

// Function to implement the game
int game(char you, char computer)
{
    // If both the user and computer
    // has choose the same thing
    if (you == computer)
        return -1;

    // If user's choice is stone and
    // computer's choice is paper
    if (you == 's' && computer == 'p')
        return 0;

            // If user's choice is paper and
            // computer's choice is stone
            else if (you == 'p' && computer == 's') return 1;

    // If user's choice is stone and
    // computer's choice is scissor
    if (you == 's' && computer == 'z')
        return 1;

    // If user's choice is scissor and
    // computer's choice is stone
    else if (you == 'z' && computer == 's')
        return 0;

    // If user's choice is paper and
    // computer's choice is scissor
    if (you == 'p' && computer == 'z')
        return 0;

    // If user's choice is scissor and
    // computer's choice is paper
    else if (you == 'z' && computer == 'p')
        return 1;
}
```

```
// Driver Code
int main()
{
    // Stores the random number
    int n;

    char you, computer, result;

    // Chooses the random number
    // every time
    srand(time(NULL));

    // Make the random number less
    // than 100, divided it by 100
    n = rand() % 100;

    // Using simple probability 100 is
    // roughly divided among stone,
    // paper, and scissor
    if (n < 33)

        // s is denoting Stone
        computer = 's';

    else if (n > 33 && n < 66)

        // p is denoting Paper
        computer = 'p';

    // z is denoting Scissor
    else
        computer = 'z';

    printf("\n\n\n\n\t\t\t\tEnter s for STONE, p for PAPER and z
for SCISSOR\n\t\t\t\t\t\t\t");

    // input from the user
    scanf("%c", &you);

    // Function Call to play the game
    result = game(you, computer);

    if (result == -1) {
        printf("\n\n\t\t\t\tGame Draw!\n");
    }
    else if (result == 1) {
        printf("\n\n\t\t\t\tWow! You have won the game!\n");
```

```
    }
    else {
        printf("\n\n\t\t\t\tOh! You have lost the game!\n");
    }
        printf("\t\t\t\tYOu choose : %c and Computer choose :
%c\n",you, computer);

    return 0;
}
```

Output:

- Firstly the user will be asked about the choice:

- When the user enters the choice then the result is displayed: